

Day 3 Switch and Router Configuration

1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Create Two VLANs in Cisco Packet Tracer

I added a **2960 switch** in Cisco Packet Tracer and created two VLANs:

- **VLAN 10 → Sales**
- **VLAN 20 → Support**

Commands

Open the switch **CLI** and enter:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Sales
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name Support
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# end
```

```
Switch# show vlan brief
```

Result

The show vlan brief command should display something similar to:

VLAN Name	Status	Ports

1 default	active	
10 Sales	active	

20 Support

active

Short Explanation

I created **VLAN 10** named **Sales** and **VLAN 20** named **Support** on the switch. VLANs logically divide a single physical switch into separate networks, allowing devices in Sales and Support to be separated even though they use the same switch.

2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Assign Switch Ports to VLANs

I assigned **three ports to VLAN 10 (Sales)** and **two ports to VLAN 20 (Support)** using the interface range command.

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# interface range fa0/1-3
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fa0/4-5
```

```
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# end
```

```
Switch# show vlan brief
```

Explanation:

Ports **Fa0/1–Fa0/3** were assigned to **Sales (VLAN 10)**, while **Fa0/4–Fa0/5** were assigned to **Support (VLAN 20)**. The show vlan brief command was used to verify the VLAN membership.

Output

```
Switch# show vlan brief
```

VLAN Name	Status	Ports

1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, ...
10 Sales	active	Fa0/1, Fa0/2, Fa0/3
20 Support	active	Fa0/4, Fa0/5

3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Connect Two Routers and Configure Static Routing

I connected **Router 1 (R1)** and **Router 2 (R2)** using an Ethernet link. The routers use 10.0.0.0/30 for the connection, while Router 2 has a LAN 192.168.2.0/24.

Router 1 Configuration

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface g0/0
```

```
Router(config-if)# ip address 10.0.0.1 255.255.255.252
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

```
Router(config)# end
```

Router 2 Configuration

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# interface g0/0
```

```
Router(config-if)# ip address 10.0.0.2 255.255.255.252
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# interface g0/1
```

```
Router(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# end
```

IP Address Plan

Device/Interface	IP Address	Subnet Mask
R1 G0/0	10.0.0.1	255.255.255.252
R2 G0/0	10.0.0.2	255.255.255.252
R2 G0/1	192.168.2.1	255.255.255.0

Verification

On Router 1, use:

```
Router# show ip route
```

The static route should appear as:

```
S 192.168.2.0/24 [1/0] via 10.0.0.2
```

The static route tells **Router 1** that traffic destined for the 192.168.2.0/24 LAN should be sent to **Router 2 at 10.0.0.2**.

4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Inter-VLAN Communication Test

I configured **Sales (VLAN 10)** and **Support (VLAN 20)** on the switch. When a device in Sales tried to ping a device in Support, the ping failed because VLANs are separate Layer 2 networks and the switch does not route traffic between them.

For example:

Sales PC: 192.168.10.2 → ping 192.168.20.2

Support PC: 192.168.20.2

The ping fails because there is **no Layer 3 device configured to route between VLAN 10 and VLAN 20**.

How to Allow Communication

Inter-VLAN routing can be configured using **Router-on-a-Stick**. The switch-to-router link is configured as a trunk, and the router uses subinterfaces:

```
Router(config)# interface g0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config)# interface g0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

The PCs would use the appropriate router subinterface as their **default gateway**. This allows the router to route traffic between **Sales VLAN 10** and **Support VLAN 20**.